

Assignment-4.3

N. Jashwanth Reddy
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Batch-46

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
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Course Code	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/I	Regulation	R23
Date and Day of Assignment	Week 2 - Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number: 3.3(Present assignment number)/24(Total number of assignments)			

Q.No.	Question	Expected Time to complete
1	Lab 4: Advanced Prompt Engineering – Zero-shot, One-shot, and Few-shot Techniques Lab Objectives - To explore and apply different levels of prompt examples in AI-assisted code generation - To understand how zero-shot, one-shot, and few-shot prompting affect AI output quality - To evaluate the impact of context richness and example quantity on AI performance	Week2 - Wednesday

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	- To build awareness of prompt strategy effectiveness for different problem types Lab Outcomes (LOs) After completing this lab, students will be able to: - Use zero-shot prompting to instruct AI with minimal context - Use one-shot prompting with a single example to guide AI code generation - Apply few-shot prompting using multiple examples to improve AI responses - Compare AI outputs across different prompting strategies	
	Task 1: Zero-Shot Prompting – Leap Year Check Scenario Zero-shot prompting involves giving instructions without providing examples. Task Description Use zero-shot prompting to instruct an AI tool to generate a Python function that: - Accepts a year as input - Checks whether the given year is a leap year - Returns an appropriate result Note: No input-output examples should be provided in the prompt. Expected Output - AI-generated leap year checking function - Correct logical conditions - Sample input and output - Screenshot of AI-generated response (if required) Code: <pre># Generate a Python function that takes a year as input and checks whether it is a leap year. '''def is_leap_year(year): if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0): return True else: return False # Example usage: year = int(input("Enter a year: ")) if is_leap_year(year): print(f"{year} is a leap year.") else: print(f"{year} is not a leap year.")'''</pre> OutPut: <pre>PS J:\AI\codes> & "C:/Program Files/Python311/python.exe" j:/AI/codes/assignment-4.py 2020 is a leap year. PS J:\AI\codes> & "C:/Program Files/Python311/python.exe" j:/AI/codes/assignment-4.py Enter a year: 2019 2019 is not a leap year.</pre>	
	Task 2: One-Shot Prompting – Centimeters to Inches Conversion Scenario One-shot prompting guides AI using a single example. Task Description Use one-shot prompting by providing one input-output example to generate a Python function that: - Converts centimeters to inches - Uses the correct mathematical formula Example provided in prompt: Input: 10 cm → Output: 3.94 inches Expected Output	

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- Python function with correct conversion logic
- Accurate calculation
- Sample test cases and outputs

Code :

```
# generate a python function that takes a input in centimeters and converts it to inches.
# Example: 1 inch = 2.54 cm

'''def cm_to_inches(cm):
    inches = cm / 2.54
    return inches
# Example usage:
cm = float(input("Enter length in centimeters: "))
inches = cm_to_inches(cm)
print(f'{cm} cm is equal to {inches} inches.'''
```

Output :

```
PS J:\AI\codes> & "C:/Program Files/Python311/python.exe" j:/AI/codes/assignment-4.py
Enter length in centimeters: 10
10.0 cm is equal to 3.937007874015748 inches.
```

Task 3: Few-Shot Prompting – Name Formatting

Scenario

Few-shot prompting improves accuracy by providing multiple examples.

Task Description

Use few-shot prompting with 2–3 examples to generate a Python function that:

- Accepts a full name as input
- Formats it as “Last, First”

Example formats:

- "John Smith" → "Smith, John"
- "Anita Rao" → "Rao, Anita"

Expected Output

- Well-structured Python function
- Output strictly following example patterns
- Correct handling of names
- Sample inputs and outputs

Code:

```
# generate a python function that takes full names as input and separates first and last names.
# Example: Input: "John Doe" Output: First Name: "John", Last Name: "Doe"
# Example: Input: "AliceMary Johnson" Output: First Name: "AliceMary", Last Name: "Johnson"

'''def separate_names(full_name):
    names = full_name.split()
    if len(names) >= 2:
        first_name = names[0]
        last_name = names[-1]
        return first_name, last_name
    else:
        return None, None
# Example usage:
full_name = input("Enter full name: ")
first_name, last_name = separate_names(full_name)
if first_name and last_name:
    print(f"First Name: {first_name}")
    print(f>Last Name: {last_name}")
else:
    print("Please enter both first and last names.'''
```

Output:

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	<pre>PS J:\AI\codes> & "C:/Program Files/Python311/python.exe" j:/AI/codes/assignment-4.py Enter full name: Nandyala Jashwanth Reddy First Name: Nandyala Last Name: Reddy PS J:\AI\codes> & "C:/Program Files/Python311/python.exe" j:/AI/codes/assignment-4.py Enter full name: Nandyala JashwanthReddy First Name: Nandyala Last Name: JashwanthReddy</pre>	
	Task 4: Comparative Analysis – Zero-Shot vs Few-Shot Scenario Different prompt strategies may produce different code quality. Task Description • Use zero-shot prompting to generate a function that counts vowels in a string • Use few-shot prompting for the same problem • Compare both outputs based on: ○ Accuracy ○ Readability ○ Logical clarity Expected Output • Two vowel-counting functions • Comparison table or short reflection paragraph • Conclusion on prompt effectiveness Code: <pre># Generate a python function that takes string as input and count the number of vowels in it. # Example: Input: "Hello World" Output: 3 # Example: Input: "Programming" Output: 3 def count_vowels(input_string): vowels = "aeiouAEIOU" count = 0 for char in input_string: if char in vowels: count += 1 return count # Example usage: input_string = input("Enter a string: ") vowel_count = count_vowels(input_string) print(f"The number of vowels in the string is: {vowel_count}")</pre> Output: <pre>PS J:\AI\codes> & "C:/Program Files/Python311/python.exe" j:/AI/codes/assignment-4.py Enter a string: jashwanth The number of vowels in the string is: 2 PS J:\AI\codes> & "C:/Program Files/Python311/python.exe" j:/AI/codes/assignment-4.py Enter a string: aeiou The number of vowels in the string is: 5</pre>	
	Task 5: Few-Shot Prompting – File Handling Scenario File processing requires clear logical understanding. Task Description Use few-shot prompting to generate a Python function that: • Reads a .txt file • Counts the number of lines in the file • Returns the line count Expected Output • Working Python file-processing function	

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- Correct line count
- Sample .txt input and output
- AI-assisted logic explanation

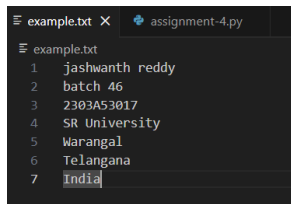
Code:

```
# Generate a python function that reads a text file and counts the number of lines in it and return the count
# Example: Input: "sample.txt" Output: 10
# Example: Input: "document.txt" Output: 25

'''def count_lines_in_file(file_path):
    try:
        with open(file_path, 'r') as file:
            lines = file.readlines()
            return len(lines)
    except FileNotFoundError:
        return "File not found."

# Example usage:
file_path = input("Enter the file path: ")
line_count = count_lines_in_file(file_path)
print(f"The number of lines in the file is: {line_count}")'''
```

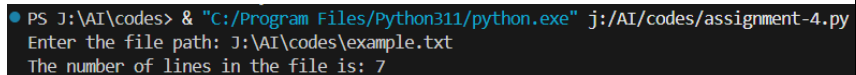
File Content:



The screenshot shows a code editor with two tabs: 'example.txt' and 'assignment-4.py'. The 'example.txt' tab is active, displaying the following text:

```
1 jashwanth reddy
2 batch 46
3 2303A53017
4 SR University
5 Warangal
6 Telangana
7 India
```

Output:



The screenshot shows a terminal window with the following output:

```
PS J:\AI\codes> & "C:/Program Files/Python311/python.exe" j:/AI/codes/assignment-4.py
Enter the file path: J:\AI\codes\example.txt
The number of lines in the file is: 7
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.